UNIVERSITY OF BRISTOL

MSC BIOINFORMATICS

Understanding the Effect of Forest Age on Machine Learning Evapotranspiration Generalisation

PROJECT PLAN

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ABSTRACT (The abstract has a maximum of 200 words)

Key words: Machine Learning, FLUXNET, Evapotranspiration.
Word count: 3000

Declaration

I declare that the work in this dissertation was carried out in accordance with the requirements of the University's Regulations and Code of Practice for Taught Programmes and that it has not been submitted for any other academic award. Except where indica ted by specific reference in the text, this work is my own work. Work done in collaboration with, or with the assistance of others, is indicated as such. I have identified all material in this dissertation which is not my own work through appropriate refer encing and acknowledgement. Where I have quoted or otherwise incorporated material which is the work of others, I have included the source in the references. Any views expressed in the dissertation, other than referenced material, are those of the author.

Archie Benn
May 3, 2026

Contents

Literature Review (3000 words)

Introduction ~800 Words

Evapotranspiration

Terrestrial evapotranspiration (ET) is a key component of the global hydrological cycle describing the transition of water from liquid to gas at Earth's surface and further transfer to the atmosphere, and is a crucial component of the energy and water exchange between terrestrial ecosystems and the climate system. The term emphasises the combination of water vapourisation flux from (1) abiotic evaporation from soil and plant canopy surfaces, and (2) biotic transpiration through stomata to the atmosphere (Katul et al., 2012). ET can be generally described through:

$$ET = T + E_{Soil} + E_{Canopy} \tag{1}$$

where T refers to transpiration through plant stomata, E_{soil} to evaporation from the soil surface, and E_{Canopy} to evaporation from plant canopies (via rainfall interception). The contribution of these variables to overall ET is not equal, with Lian et al. (2018) estimating a global T/ET ratio of 0.62±0.06, indicating plant transpiration as the leading component of ET worldwide. The partitioning of ET into E ($E_{soil} + E_{Canopy}$) and T within ecosystems is associated with "water use efficiency" and has use in irrigation management, where T is linked with plant productivity, while E does not directly contribute to production (Kool et al., 2014). This in turn likely explains the elevated E/ET ratios in managed croplands compared to natural ecosystems (Yang et al., 2023). Accurate partitioning estimates are also important in aiding carbon cycle projections given the coupled effect of transpiration and land-atmosphere carbon dioxide exchanges through stomata.

The Complexities of Ageing

The magnitude of ET is governed by many environmental factors such as solar radiation, leaf area index (LAI), air temperature, vapour pressure deficit (VPD), soil moisture, and wind speed (Yang et al., 2023). Of these, Lian et al. (2018) identified LAI as the primary driver of T/ET spatial patterns, with Kool et al. (2014) suggesting that in systems with full canopy cover (and therefore high LAI) ET is often assumed to be "similar enough to T to permit correlation of ET with biomass". However, this assumption may break down in densely vegetated systems with multi-layer canopies such as forests, despite high LAI. In these systems, the 2-dimensional remote measure of LAI will not capture the vertical canopy complexity which drives rainfall interception and subsequent E_{Canopy}, implying that canopy cover indices may struggle to explain T/ET ratios in more complex vertical environments. This is reflected by Zhou et al. (2016) who report higher coefficients of determination between enhanced vegetation index (EVI) and T/ET in corn and soybean sites (0.84 and 0.82) with simple canopies, compared to evergreen needle leaf forests (0.37), a result possibly due to increased E_{Canopy} driving this ratio down.

Oishi et al. (2020), in a chronosequence study on broadleaf forests in the US, report that interception of incoming rainfall strongly increases with age from ~2% (12 year old stand) to 20% (200 year old stand) despite only small differences in LAI, an effect in part due to branch architecture which varies with age. This study also reported that annual canopy transpiration (T) was over twice as high in the 35 year old stand compared to the 12 and 200 year old stands, with increasing canopy interception in the oldest stand largely offsetting the reduction in T such that total ET remained similar between the two older stands, but much lower in the youngest stand (Fig. 2). Furthermore, the high LAI across all stand ages intercepted much of the incoming radiation before it reached the forest floor, revealing a relationship between LAI and E_{Soil} but not necessarily E_{Canopy} in these densely vegetated areas. This study demonstrates how age driven effects in canopy architecture introduce complexity and affect overall ET, as well as the relative contributions of T and E_{Canopy} which may vary significantly over time within stands. While vegetation indices such as LAI may capture ET dynamics in sparsely vegetated or simple landscapes, this relationship breaks down in ecosystems where age driven changes to ET and ET partitioning through variations in canopy structure introduce complexity that LAI cannot resolve.

Alongside canopy architecture, stomatal conductance (g_s) declines as trees grow taller due to gravitational reductions in leaf water potential and decreasing hydraulic conductance with increasing path length from soil to leaf (Ryan and Yoder, 1997). For example, Schäfer et al. (2000) demonstrated that the mean stomatal conductance of tree crowns decreased by approximately 60% over a 30m increase in tree height, and Hubbard et al. (1999) established a 44% drop in hydraulic conductance in old trees compared to young trees in a mixed age stand. As g_s directly influences T (Monteith, 1965), these age related changes introduce further complexity into ET dynamics and suggest that age could be a key driver of ET in forest ecosystems.

Climate Change and Forest Age

Forest age distributions are increasingly being affected by climate change-driven disturbances such as fire, drought, insects, and pathogens (Seidl et al., 2017), with these expected to continue in the coming decades (Solomon et al., 2009).

The forest carbon balance in regions with newly disturbed and old-growth forests may not be realistically estimated without explicitly constraining data-driven statistical models with a disturbance history or forest demography. (Mapping global forest age from forest inventories, biomass and climate data)

"Forest age distributions are highly dynamic, shaped by harvesting, fire, storm damage, and increasingly by climate change driven disturbance regimes (**?**). As global forests trend towards greater age heterogeneity — with large areas of young regrowing forest alongside ageing secondary stands — the age-driven complexity in ET dynamics described above has growing implications for regional water and carbon cycle projections. Yet whether these differences are sufficient to limit the generalisation of data-driven ET models across age classes remains unexplored."

Evapotranspiration Prediction ~1400 Words

(What don't we know and what are the issues?)

Contemporary ET Prediction Methods

FLUXNET

Machine Learning Models for ET Prediction

Machine Learning Applicability ~800 Words

(What is the case for using ML here?)

Why use Machine Learning?

Machine Learning Generalisation?

Project Aims and Objectives (500 words)

Aims

- Overview of how this project will answer the question/area proposed at end of introduction.

Project Plan (500 words)

Data Acquisition

Site Selection

Analysis

Feature Engineering

ML Model Setup and Training

ML Model Evaluation

ML Model Predictions

Tools

Interpretation

Gantt Chart

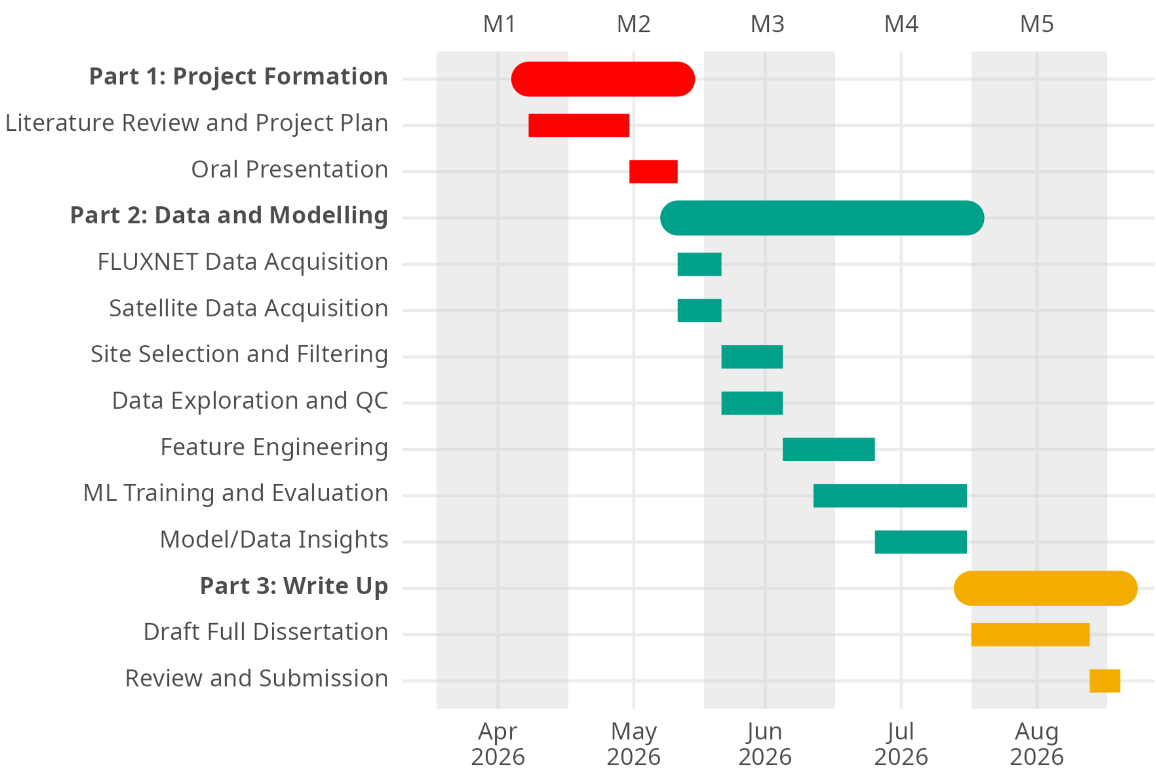

Figure 1: A proposed Gantt Chart for the duration of this project.

Risk Register (~ 250 words)

References

Hubbard, R. M., Bond, B. J., and Ryan, M. G. (1999). Evidence that hydraulic conductance limits photosynthesis in old pinus ponderosa trees. *Tree physiology*, 19(3):165–172.

Katul, G. G., Oren, R., Manzoni, S., Higgins, C., and Parlange, M. B. (2012). Evapotranspiration: A process driving mass transport and energy exchange in the soil-plant-atmosphere-climate system. *Reviews of Geophysics*, 50(3).

Kool, D., Agam, N., Lazarovitch, N., Heitman, J., Sauer, T., and Ben-Gal, A. (2014). A review of approaches for evapotranspiration partitioning. *Agricultural and forest meteorology*, 184:56–70.

Lian, X., Piao, S., Huntingford, C., Li, Y., Zeng, Z., Wang, X., Ciais, P., McVicar, T. R., Peng, S., Ottlé, C., et al. (2018). Partitioning global land evapotranspiration using cmip5 models constrained by observations. *Nature Climate Change*, 8(7):640–646.

Monteith, J. L. (1965). Evaporation and environment.

Oishi, A., Denham, S., Brantley, S., Novick, K., Bolstad, P., and Miniat, C. (2020). Quantifying the effects of stand age on components of forest evapotranspiration. *Acta Horticulturae*, 1300:89–96.

Ryan, M. G. and Yoder, B. J. (1997). Hydraulic limits to tree height and tree growth. *Bioscience*, 47(4):235–242.

Schäfer, K., Oren, R., and Tenhunen, J. (2000). The effect of tree height on crown level stomatal conductance. *Plant, Cell & Environment*, 23(4):365–375.

Seidl, R., Thom, D., Kautz, M., Martin-Benito, D., Peltoniemi, M., Vacchiano, G., Wild, J., Ascoli, D., Petr, M., Honkaniemi, J., et al. (2017). Forest disturbances under climate change. *Nature climate change*, 7(6):395–402.

Solomon, S., Plattner, G.-K., Knutti, R., and Friedlingstein, P. (2009). Irreversible climate change due to carbon dioxide emissions. *Proceedings of the national academy of sciences*, 106(6):1704–1709.

Yang, Y., Roderick, M. L., Guo, H., Miralles, D. G., Zhang, L., Fatichi, S., Luo, X., Zhang, Y., McVicar, T. R., Tu, Z., et al. (2023). Evapotranspiration on a greening earth. *Nature Reviews Earth & Environment*, 4(9):626–641.

Zhou, S., Yu, B., Zhang, Y., Huang, Y., and Wang, G. (2016). Partitioning evapotranspiration based on the concept of underlying water use efficiency. *Water Resources Research*, 52(2):1160–1175.

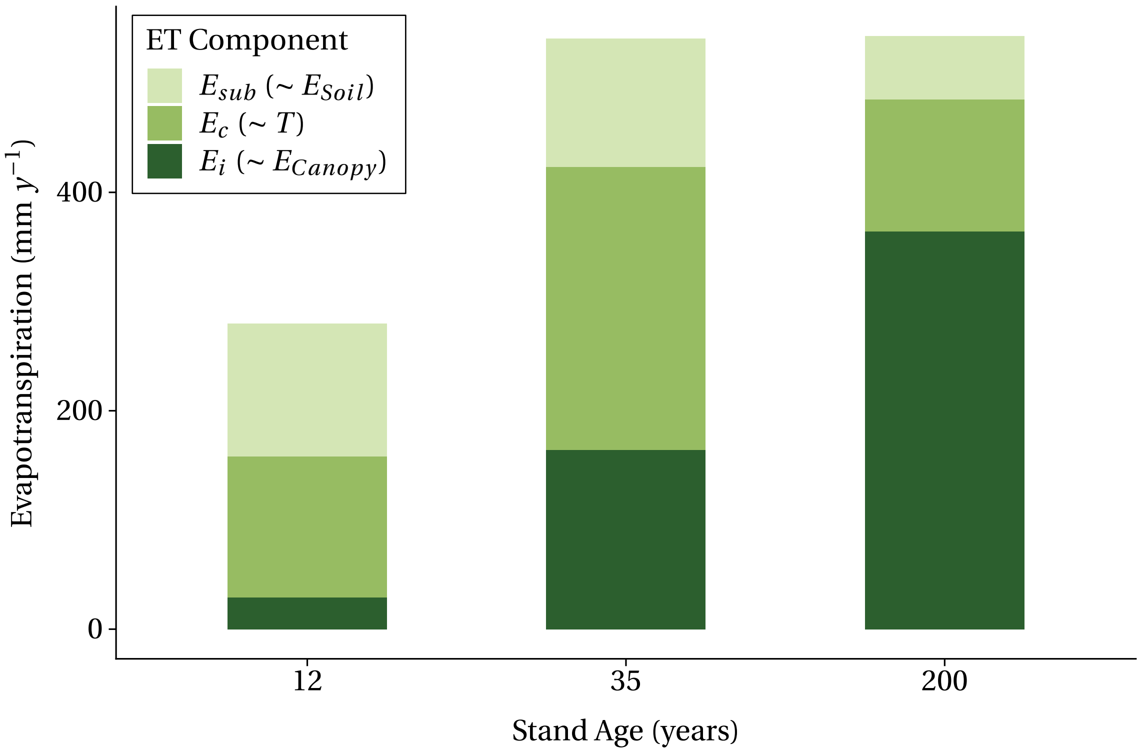

Figure 2: Components of annual evapotranspiration by stand age, redrawn from Oishi et al. (2020). Sub canopy evapotranspiration (E_{sub}), canopy transpiration (E_c), and canopy interception (E_i) correspond approximately to E_{Soil}, T, and E_{Canopy} in Equation 1, although E_{sub} additionally captures under canopy transpiration.

Method	Advantages	Disadvantages